

Non-Market Valuation and Regulation

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Non-Market Valuation and Regulation

Outline for Today: Non-Market Valuation as the Fundamental Problem of Environmental Economics

What are the benefits of improving environmental quality?

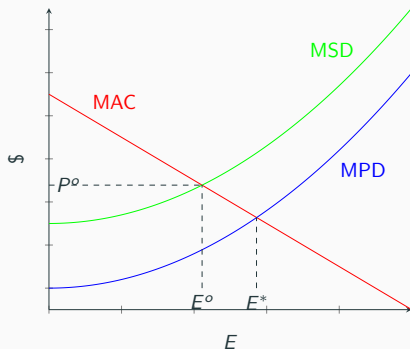
- Ask people (stated preferences)
- Watch people (revealed preferences)
 - Hedonics and Recreation Demand
- Bottom up accounting
 - Add up observable physical impacts and apply a dollar value (crop yields, mortality)

How should we aggregate benefits across different (groups of) people?

- Distributional considerations and environmental justice

The Fundamental Problem of Environmental Economics

Last time - Efficient Pigovian taxes require knowledge of the magnitude of marginal external costs (e.g. Social Cost of Carbon)



How do we know the slope of the damage function? Unlike private goods, no prices we can use.

Cost-Benefit Analysis

This is all basically a way of saying we are doing cost-benefit analysis (your assignment for this class!)

Many regulations can be modelled as some sort of tax

- We want to make sure the marginal benefits (reduced social damages) outweigh the marginal costs (abatement)

But a key question in environmental economics is how to value non-monetary benefits

- Better health from cleaner air or water
- Recreation opportunities
- Existence values

Story Time: The Tellico Dam and the Snail Darter



Story Time: The Tellico Dam and the Snail Darter



TRILOBITES

This Tiny Fish's Mistaken Identity Halted a Dam's Construction

Scientists say the snail darter, whose endangered species status delayed the building of a dam in Tennessee in the 1970s, is a genetic match of a different fish.

Stated Preferences

Stated Preferences

Simple idea: How much would you be willing to pay to save the snail darter?

- Economists tend to be skeptical of hypothetical questions.
- Under fairly weak assumptions looking at how people make choices in the world reveals their true preferences.

Carson and Groves (2007): under the right circumstances, responses to stated preference surveys can be treated as revealed preferences:

- Need to believe that answers will be used to inform real policy.
- Need to believe there is some mechanism (taxes) that they will actually enforce payment.
- In other words, it needs to seem as if the stakes are real.

Exxon Valdez Oil Spill 1989



Exxon Valdez Oil Spill 1989



When the Exxon Valdez ran aground on Bligh Reef on March 24, 1989, it was transporting 53 million gallons of Alaskan North Slope crude oil bound for California. The spill dumped 10.8 million gallons of the tanker's cargo into Prince William Sound, eventually damaging 1,300 miles of shoreline. Staff/TNS/Newscom

Passive Use vs Direct Use Values

- Hausman, Leonard, McFadden (1995) found \$4 million lost recreation value
- Exxon settled with fishermen for \$67 million

Passive Use vs Direct Use Values

- Hausman, Leonard, McFadden (1995) found \$4 million lost recreation value
- Exxon settled with fishermen for \$67 million
- Carson et al (1992) found passive use values of \$3 *billion*

Contingent Valuation Survey

Here's how the program would work.

Two large Coast Guard ships specially designed for Alaskan waters will escort each tanker from Valdez all the way through Prince William Sound until they get to the open sea. These escort ships will do two things.

First, they will help prevent an accident in the Sound by making it very unlikely that a tanker will stray into dangerous waters. (PAUSE)

Second, if an accident does occur, the escort ships will carry the trained crew and special equipment necessary to keep even a very large spill from spreading beyond the tanker. (PAUSE)

This drawing shows how this would be done. (PAUSE)

SHOW CARD 6

Escort ship crew would immediately place a boom that stands four feet above the water and five feet below the water, called a Norwegian sea fence, around the entire area of the spill. (POINT IF NECESSARY) Because oil floats on the water, in the first days of a spill, the sea fence will keep it from floating away. The oil trapped by the sea fence would be scooped up by skimmers, and pumped into storage tanks on the escort ships. Within hours, an emergency rescue tanker would come to the scene to aid in the oil recovery and transport the oil back to Valdez.

This system has been used successfully in the North Sea by the Norwegians.

Contingent Valuation Survey

Because two tankers usually sail from Valdez each day, the Coast Guard would have to maintain a fleet of escort ships, skimmers, and an emergency tanker, along with several hundred Coast Guard crew to run them.

Although the cost would be high, the escort ship program makes it virtually certain there would be no damage to Prince William Sound's environment from another large oil spill during the ten years it will take all the old tankers to be replaced by double-hulled tankers.

It is important to note that this program would not prevent damage from a spill anywhere else in the United States because the escort ships could only be used in Prince William Sound.

If the program was approved, here is how it would be paid for.

All the oil companies that take oil out of Alaska would pay a special one time tax which will reduce their profits. Households like yours would also pay a special one time charge that would be added to their federal taxes in the first year and only the first year of the program.

This money will go into a Prince William Sound Protection Fund. The one time tax will provide the Fund with enough money to pay for the equipment and ships and all the yearly costs of running the program for the next ten years until the double hulled tanker plan takes full effect. By law, no additional tax payment could be required.

Contingent Valuation Survey

Because everyone would bear part of the cost, we are using this survey to ask people how they would vote if they had the chance to vote on the program.

We have found some people would vote for the program and others would vote against it. Both have good reasons for why they would vote that way.

Those who vote for say it is worth money to them to prevent the damage from another large spill in Prince William Sound.

Those who vote against mention concerns like the following.

Some mention that it won't protect any other part of the country except the area around Prince William Sound.

Some say that if they pay for this program they would have less money to use for other things that are more important to them.

And some say the money they would have to pay for the program is more than they can afford.

(PAUSE)

Contingent Valuation Survey

Of course whether people would vote for or against the escort ship program depends on how much it will cost their household.

At present, government officials estimate the program will cost your household a total of \$60. You would pay this in a special one time charge in addition to your regular federal taxes. This money would only be used for the program to prevent damage from another large oil spill in Prince William Sound. (PAUSE)

If the program cost your household a total of \$60 would you vote for the program or against it?

Endangered Species: Loomis and White (1996)

Table 1
Summary of economic values of rare and T/E species (\$1993)

	Low value	High value	Average of all studies
<i>Studies reporting annual WTP</i>			
Northern spotted owl	\$44	\$95	\$70
Pacific salmon/Steelhead	\$31	\$88	\$63
Grizzly bears			\$46
Whooping cranes			\$35
Red-cockaded woodpecker	\$10	\$15	\$13
Sea otter			\$29
Gray whales	\$17	\$33	\$26
Bald eagles	\$15	\$33	\$24
Bighorn sheep	\$12	\$30	\$21
Sea turtle			\$13
Atlantic salmon	\$7	\$8	\$8
Squawfish			\$8
Striped shiner			\$6
<i>Studies reporting lump sum WTP</i>			
Bald eagles	\$178	\$254	\$216
Humpback whale			\$173
Monk seal			\$120
Gray wolf	\$16	\$118	\$67
Arctic grayling/Cutthroat trout	\$13	\$17	\$15

WTP for 2 degrees climate change: Kotchen and Ashenfarb (2023)

Table 1: Summary of responses to the WTP question for each specified dollar amount

Specified amount	Total obs.	Yes	No	Protest responses	Refused responses
\$6	101	56	12	26	7
\$16	102	39	20	39	4
\$26	202	95	47	56	4
\$45	203	88	62	47	6
\$85	204	75	70	49	10
\$124	103	30	32	38	3
\$165	100	22	45	30	3
Total	1015	405	288	285	37

Criticisms and Caveats

- Hypothetical biases
 - Might not know, might want to please the researcher, might be cheap talk

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- Scope Insensitivity (Desvousges et al 1992)

Table 5-1. Univariate Results for Three Migratory-Waterfowl Questionnaires: Censored Data Sets

	2,000 Birds (WTP \$)	20,000 Birds (WTP \$)	200,000 Birds (WTP \$)
Mean	80	78	88
Standard Deviation	187	132	166
Median	25	25	25
Mode	0	100	100
Range	0-1,550	0-1,000	0-1,000
Shapiro-Wilk statistic ^a	0.43	0.60	0.54
N	288	286	281

^a This test statistic indicates that these distributions are not normal.

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 - Might not know, might want to please the researcher, might be cheap talk
- Framing effects
 - 1/3 will live vs 2/3 will die (Kahneman and Tversky 1981).
- Scope Insensitivity (Desvousges et al 1992)
- Endowment effects
 - Divergence between WTP and WTA

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Even if CV isn't perfect, it is clearly better than the alternative of assigning zero to existence values

- Clearly there are also issues with revealed preference assumptions
- CV compares favorably with similarly worded ballot initiatives

A properly designed survey is of the highest importance

Revealed Preferences

Revealed Preferences and Close Complements

Sometimes we can infer something about the value of a non-market good by studying markets for closely related private goods

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- Easy to put a price on whale watching tours

Revealed Preferences and Close Complements

Sometimes we can infer something about the value of a non-market good by studying markets for closely related private goods

- Hard to put a price on whales (no market)
- Easy to put a price on whale watching tours
- Price of the latter can tell us something about the value of the former
 - Recreation demand or travel cost models estimate the expenditures induced by a marginal change in the environmental good as a lower-bound on the value of the good.

When people buy a house they are purchasing a bundle of goods:

- Roof and walls

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Hedonic Methods: Housing

When people buy a house they are purchasing a bundle of goods:

- Roof and walls
- School district
- Environmental quality

Real Estate markets have several other useful features:

- Competitive and thick
- Consequential - buyers and sellers likely to be well-informed
- Clearly reflect the value of local public goods

The Hedonic Model

House prices are a function of characteristics: $P(x, E)$:

- x : rooms, bathrooms, school district, distance to boulangerie
- E : local air quality, water quality, park, etc...
- Prices arise in equilibrium of buyer-seller interactions

Consumers choose a house to solve:

$$\max_{x, E, z} u(x, E, z) \quad (1)$$

$$y = z + P(x, E)$$

Assumes consumers can choose E (by sorting across space?)

- z is purchases on all other goods, y is income

The Hedonic Model

$$\max_{x,E,z} U(x, E, z) + \lambda(y - z - P(x, E)) \quad (2)$$

First order conditions give:

$$\frac{dU}{dx} = \lambda \frac{dP}{dx}$$

$$\frac{dU}{dE} = \lambda \frac{dP}{dE}$$

$$\frac{dU}{dz} = \lambda$$

Rearranging the last two gives:

$$\frac{\frac{dU}{dE}}{\frac{dU}{dz}} = \frac{dP}{dE}$$

The Hedonic Model

The left hand side is the MRS between environmental goods and other goods: willingness to trade of other purchases for E:

$$\frac{\frac{dU}{dE}}{\frac{dU}{dz}} = \frac{dP}{dE}$$

The right hand side is how a marginal change in E changes the price of the house. In principle, we can estimate this in a regression:

$$\log P_i = \beta_1 x_i + \beta_2 E_i + e_i \quad (3)$$

$$\frac{d}{dE} \log P = \beta = \frac{\frac{dP}{dE}}{P} \quad (4)$$

The Hedonic Method: Identification

$$\log P_i = \beta_1 x_i + \beta_2 E_i + e_i \quad (5)$$

Why doesn't this work in practice?

- e_i could be correlated with E and P_i
- x_i s are endogenous, could have been chosen as a result of E .
- We need exogenous variation in E

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Chay and Greenstone (JPE) 2003: Look at changes in home prices resulting from 'exogenous' changes in air pollution

- In 1975, the Clean Air Act designated certain counties as non-attainment if they were above a pollution threshold
- In principle counties just below and just above that threshold should be very similar, except that areas just above the threshold experienced bigger improvements in air quality

The Hedonic Method: Results

People are willing to pay for better AQ (about 2% per $1\mu\text{g}$).

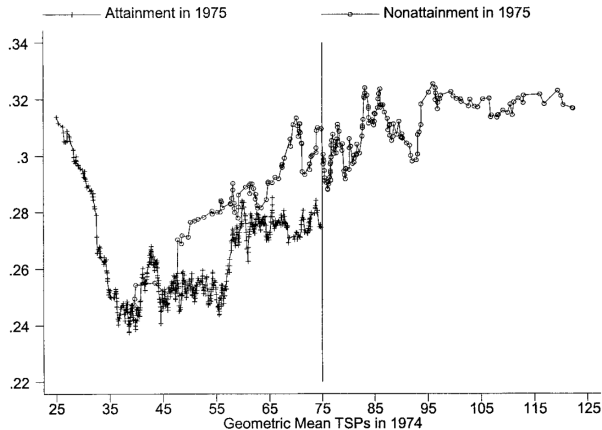


FIG. 5.—1970–80 change in log housing values by 1975 nonattainment status and the geometric mean of TSPs in 1974.

Takeaways

Hedonics are often the first tool in the kit for environmental economists

- Muehlenbachs, Spiller and Timmins (2015) - Fracking
- Schlenker et al (2005) - Agriculture and Global Warming
- Kaiser and Shapiro (2018) - Clean Water Act - municipal wastewater treatment

Several important considerations:

- Only captures use values related to the property
- Mobility between houses
 - See Christensen and Timmins (2022). Sorting or Steering: The Effects of Housing Discrimination on Neighborhood Choice
- Perfect information
- Well functioning capital markets

Design a stated-preference and a revealed preference approach to study the value of:

- The Mona Lisa
- Scientific research on a vaccine for cancer
- Honeybees
- Better weather forecasts
- Bringing high speed internet to a rural village

What are some limitations of your studies?

Bottom up

Estimating Pollution Damages: Bottom Up

Given the assumptions involved in both revealed and stated preference approaches, empirical researchers often attempt to measure the effect of pollution on more clearly defined physical variables:

- Crop Yields
- Morbidity and Mortality
- Property damages of natural disasters

Improvement in understanding of causal inference contributing to more credible identification. What is the problem with:

$$Y_i = \beta PM2.5_i + e_i \quad (6)$$

The Mortality and Medical Costs of Air Pollution: Evidence from Changes in Wind Direction

Tatyana Deryugina

Garth Heutel

Nolan H. Miller

David Molitor

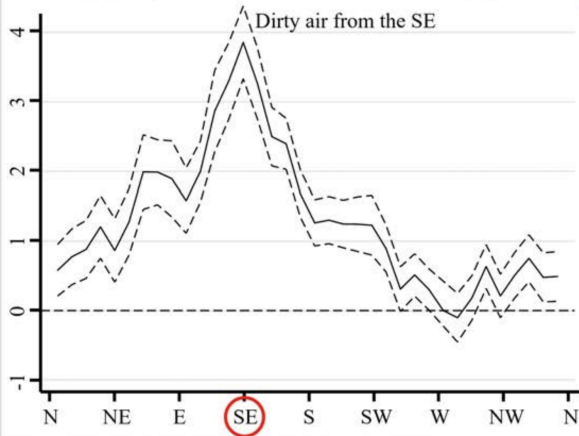
Julian Reif

AMERICAN ECONOMIC REVIEW
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(pp. 4178-4219)

Wind Direction as an IV

San Francisco, CA regional wind direction and pollution

PM 2.5 ($\mu\text{g}/\text{m}^3$) relative to wind from the West (W)



Regression Specification:

$$Y_{cdmy} = \beta PM2.5_{cdmy} + X_{cdmy}\gamma + \alpha_c + \alpha_s m + \alpha_{my} + e_{cdmy} \quad (7)$$

- Y_{cdmy} is outcome in county c on day d in month m and year y : three-day total death rate ($d, d+1, d+2$) per million, hospital admissions
- daily PM2.5 levels + 2 leads + 2 lags
- High dimensional controls: daily max temp into 17 bins, same for min temp, indicators for deciles of daily precipitation and wind speed, indicators for all possible interactions of these temp, precipitation, and wind speed variables (28,899)
- County c , state-by-month, and month-by-year FE
- Cluster se at the county level and weight

First stage:

$$PM2.5_{cdmy} = \sum_g \sum_{b=0}^2 WindDir_{cdmy}^{90b} + X_{cdmy}\sigma + \alpha_c + \alpha_s m + \alpha_{my} + e_{cdmy} \quad (8)$$

- $WindDir_{cdmy}^{90b} = 1$ if daily average wind direction in county c falls in $[90b, 90B + 90]$ and 0 otherwise
- 100 spatial group g for pollution monitors

Wind Direction as an IV: Results

Table 2: OLS and IV estimates of effect of PM 2.5 on elderly mortality, by age group

	(1) 65+	(2) 65–69	(3) 70–74	(4) 75–79	(5) 80–84	(6) 85+
Panel A: OLS estimates						
PM 2.5 ($\mu\text{g}/\text{m}^3$)	0.095*** (0.021)	0.041*** (0.014)	0.029 (0.018)	0.022 (0.022)	0.142*** (0.036)	0.425*** (0.072)
Dep. var. mean	385	131	197	312	508	1,127
Effect relative to mean, percent	0.025	0.032	0.015	0.007	0.028	0.038
Observations	1,980,549	1,980,549	1,980,549	1,980,549	1,980,549	1,980,549
Adjusted R-squared	0.254	0.080	0.085	0.082	0.077	0.110
Panel B: IV estimates						
PM 2.5 ($\mu\text{g}/\text{m}^3$)	0.685*** (0.061)	0.267*** (0.066)	0.329*** (0.068)	0.348*** (0.098)	0.877*** (0.159)	2.419*** (0.246)
F-statistic	298	285	292	303	309	315
Dep. var. mean	385	131	197	312	508	1,127
Effect relative to mean, percent	0.178	0.204	0.167	0.111	0.173	0.215
Observations	1,980,549	1,980,549	1,980,549	1,980,549	1,980,549	1,980,549

Notes: Table reports OLS and IV estimates of equation (1) from the main text. Dependent variable is the three-day mortality rate per million beneficiaries in the relevant age group. All regressions include county, month-by-year, and state-by-month fixed effects; flexible controls for temperatures, precipitation, and wind speed; and two leads of these weather controls. OLS (IV) estimates also include two lags and two leads of PM 2.5 (instruments). Estimates are weighted by the number of beneficiaries in the relevant age group. Standard errors, clustered by county, are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

How big are these effects?

How can we compare these effects to the costs of reducing air pollution? We need a common denominator

Value of Statistical Life (VSL) approach:

- Several ways of estimating WTP for reducing the probability of death:
 - wage-premia on risky jobs (hedonic wage regressions)
 - WTP for safety features on e.g. cars, medicines that extend lifespan
 - contingent valuation surveys...

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- Back to relying on revealed preference or CV!
- Equity issues can get very difficult: less money = less WTP
 - Bressler and Heal (2022): should we value 1 life in France = 82 lives in DRC?
 - We don't use different VSL for poor vs rich within France
 - Either equity weighting or global average VSL?

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Cost-effectiveness approach: Cost per life saved

- Is this the most cost-effective way to save lives?
- Benchmark: Against Malaria Foundation \$5000/life

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But there is another problem as well...

- Estimates are static, historical, don't account for adaptation

Effects of Temperature Shocks on Mortality

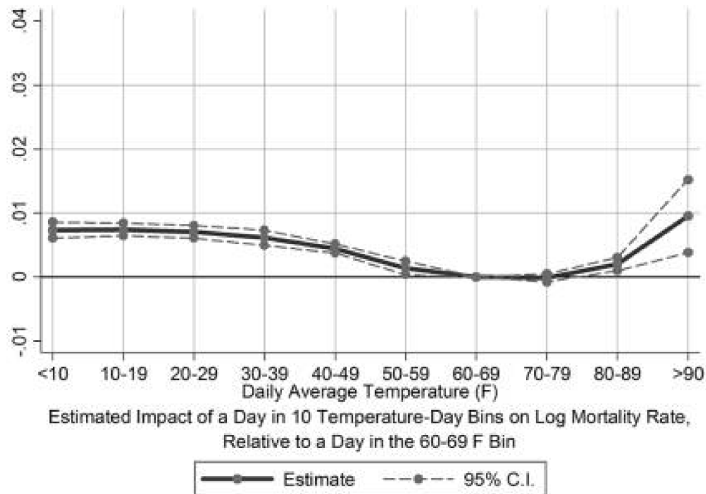
Barreca et al (2016):

$$\log(Y_{sym}) = \sum_j \theta_j T_{symj} + X_{sym}\beta + \alpha_{sm} + \rho_{ym} + e_{ysm} \quad (9)$$

- Semi-parametric approach to temperature - number of days in a month in a certain degree range
- Time varying controls for precipitation and population age structure
- State seasonal fixed effects and national month fixed effects

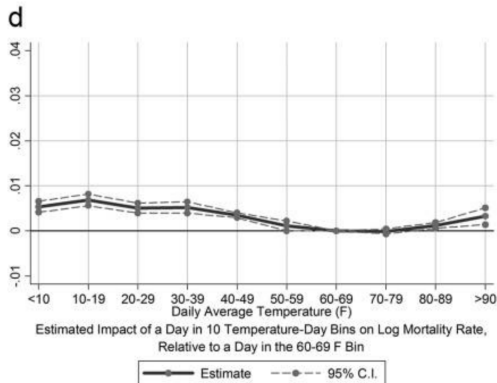
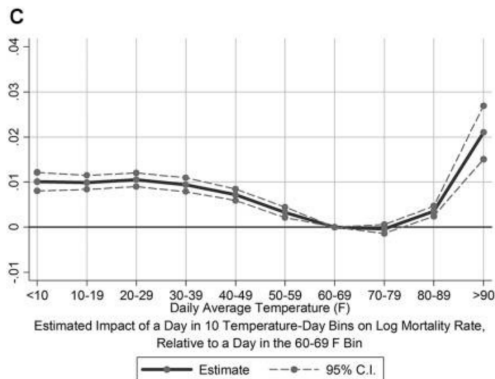
Barecca et al Results

High and Low Temps Increase Mortality



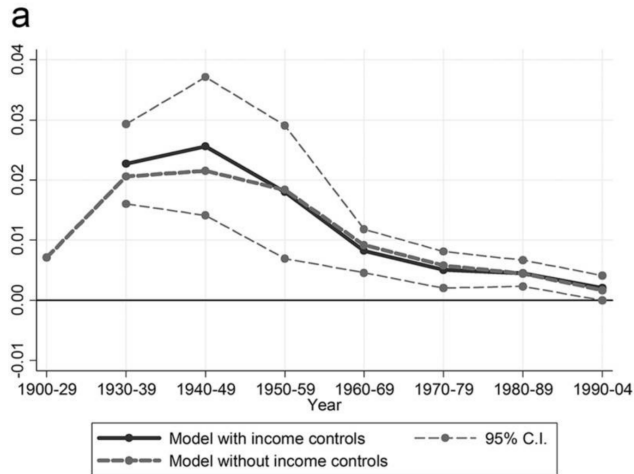
Barecca et al Results

On the left - effects before 1960, on the right - after 1960



Barecca et al Results

The effect of a hot day on mortality over time

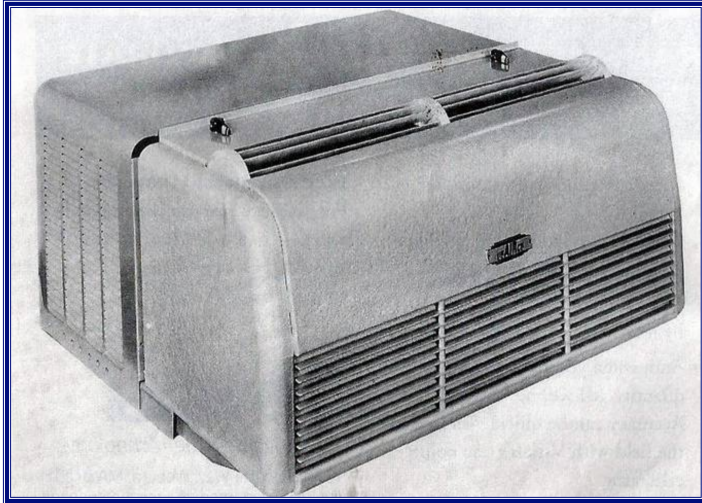


Adaptation and Innovation



De La Vergne room air conditioning unit, mid-1930s

Adaptation and Innovation



Window air conditioning unit by US Air Conditioning Corporation, c.1950

TABLE 8
ROBUSTNESS ANALYSIS OF THE EFFECT OF RESIDENTIAL AIR CONDITIONING
ON THE TEMPERATURE-MORTALITY RELATIONSHIP, 1960–2004

	(1)	(2)	(3)	(4)	(5)
Number of days above 90°F × share with residential AC	−.0212*** (.0054)	−.0212*** (.0055)	−.0343* (.0139)	−.0376*** (.0065)	−.0264** (.0088)
Number of days between 80°F and 89°F × share with residential AC	−.0048*** (.0010)	−.0048*** (.0010)	−.0060** (.0020)	−.0041** (.0013)	−.0013 (.0011)
Number of days below 40°F × share with residential AC	−.0004 (.0009)	−.0003 (.0009)	.0038 (.0024)	.0016 (.0014)	−.0010 (.0012)
Baseline controls	Yes	Yes	Yes	Yes	Yes
State-month cubic time trends	No	Yes	No	No	No
2-year window around census years	No	No	Yes	No	No
Temperature × year trends	No	No	No	Yes	No
Exposure window = 4 months	No	No	No	No	Yes
Observations	26,411	26,411	4,655	26,411	26,313

Takeaways

- Bottom up approaches have increased in popularity due to advances in measurement and causal inference
 - Biden administration moved to a bottom-up approach to measure the Social Cost of Carbon
- Still need to rely on assumptions to get something useful for policy
- Lucas Critique: Can be tricky to project econometric results into the future

Comparing Policies

Marginal Value of Public Funds

Return to our Social Welfare Function

$$W = \sum_i \alpha_i U_i(E) \quad (10)$$

$$\text{such that } \mathbf{B} \geq B(E). \quad (11)$$

α are the weights we place on each individual's utility and $B(E)$ is the spending on the environmental program.

$$\frac{dW}{dE} = \sum_i \alpha_i \frac{dU_i}{dE}. \quad (12)$$

Marginal Value of Public Funds

We never observe $\frac{dU_i}{dE}$. But we do (sometimes) observe marginal willingness to pay:

$$WTP_i = \frac{\frac{dU_i}{dE}}{\frac{dU_i}{dx_i}} \quad (13)$$

Rewrite:

$$\frac{dW}{dE} = \sum_i \alpha_i \frac{dU_i}{dE} = \sum_i \alpha_i \frac{dU}{dx_i} WTP_i. \quad (14)$$

Now define:

$$\eta = \frac{\sum_i \alpha_i \frac{dU_i}{dx_i} WTP_i}{\sum_i WTP_i}. \quad (15)$$

This is a constant, and summarizes normative considerations.

Marginal Value of Public Funds

Rewrite again, and normalize by spending:

$$\frac{\frac{dW}{dE}}{\frac{dB}{dE}} = \eta \frac{\sum_i WTP_i}{dB/dE} = \eta MVPF. \quad (16)$$

Note dB/dE includes any 'fiscal externalities'

- Spend \$1 on a policy to improve health
- Govt spending on healthcare declines

Can an MVPF be infinite?

Marginal Value of Public Funds

Should we spend an additional dollar on program A or B? We prefer A iff:

$$\frac{\eta^A}{\eta^B} > \frac{MVPF^B}{MVPF^A} \quad (17)$$

Roughly, if $\frac{MVPF_B}{MVPF_A} = 2$, we should only prefer A if we prefer the beneficiaries of A receiving \$1 more than the beneficiaries of B receiving \$2.

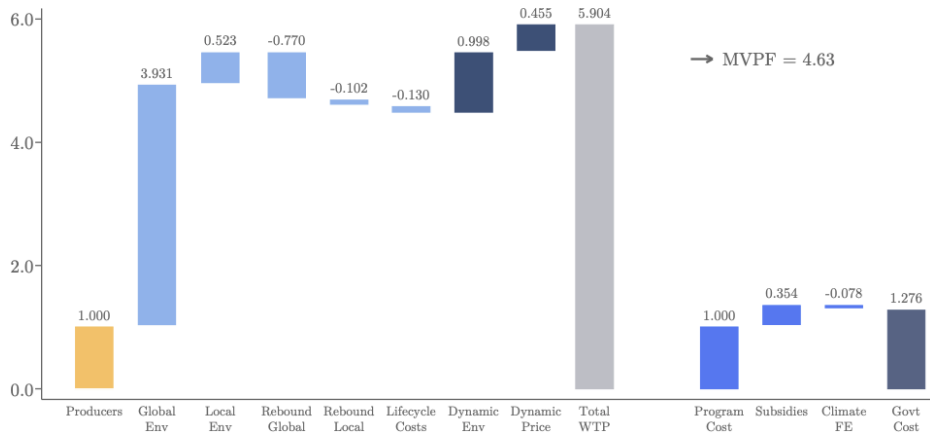
A Welfare Analysis of Policies Impacting Climate Change

Working paper by Hahn, Hendren, Metcalfe, and Sprung-Keyser

- Quantify the MVPF of a variety of policies to address climate change
- Important: not just counting environmental benefits, but all benefits
- Requires estimates of effects of policies, but also monetary values of non-market benefits, dynamic effects, and fiscal externalities
- Disaggregate by different groups to allow policy makers to assess distributional concerns

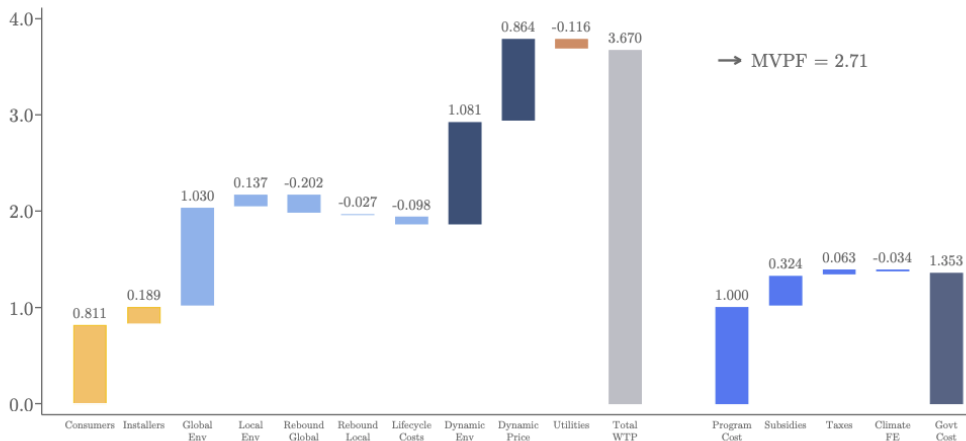
A Welfare Analysis of Policies Impacting Climate Change

A. Baseline Estimates from Hitaj (2013)

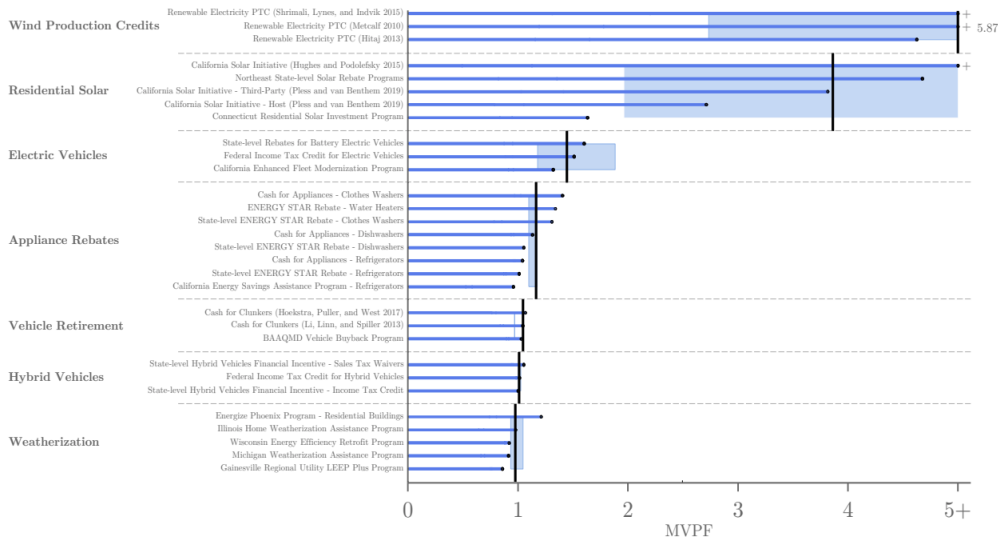


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A. Baseline Estimates from Pless and Van Bentham (2019)

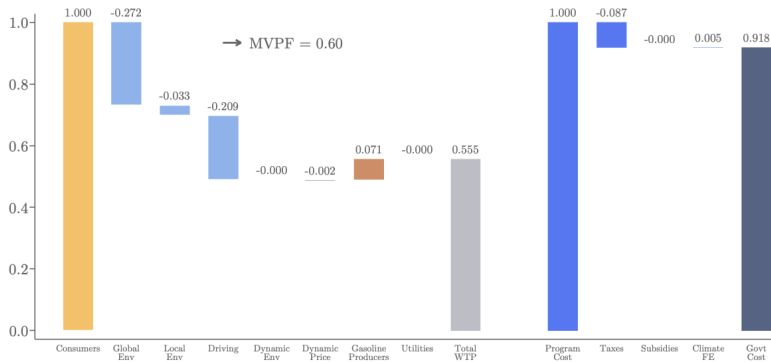


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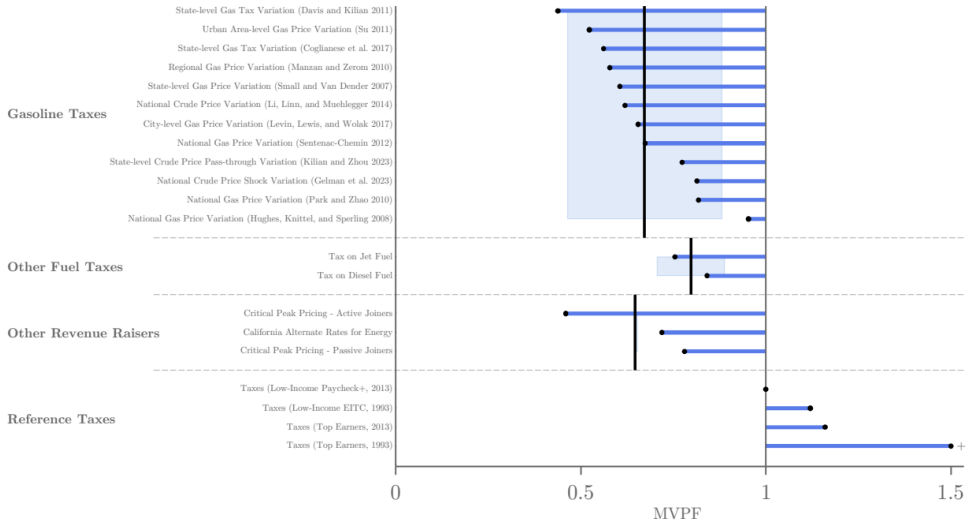


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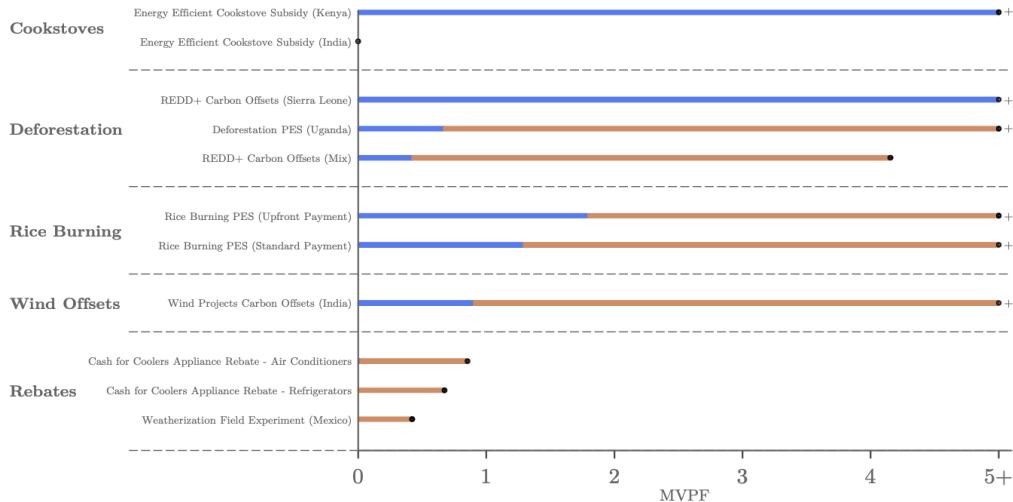
FIGURE 7: MVPF of a Gasoline Tax
Baseline Estimates from Small & Van Dender (2007)



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Equity

Social Choice and Interpersonal Comparisons

Typical CBA ignores the α_i , implicitly gives equal weight to all individuals denominated in dollars: a dollar to person A equals a dollar to person B:

$$W = \sum_i \alpha_i (b_i(E) - c_i(E)) \quad (18)$$

This is consistent with Kaldor Hicks: If we maximize the number of dollars created, we can arrange some side payments such that everyone would be better off

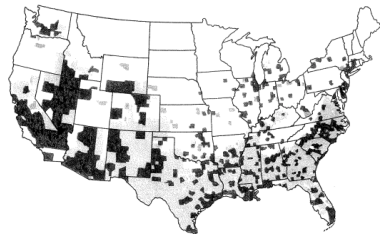
- But these transfers rarely seem to occur in practice
- Discounting and risk aversion both rely on concave utility functions: diminishing marginal returns to a dollar
 - Implies a dollar to a poor person worth more than a dollar to a rich person
- New US govt guidance opens the door to 'equity weighting' in CBA: $\alpha_i = U'(x_i)$
 - Implies individuals have the same utility functions

US Environmental Justice Movement

- 1982 Warren County Hazardous Waste Protests
- 1987 United Church of Christ Report
- Continuing documented disparities in a range of pollutants

TOXIC WASTES AND RACE In The United States

**A National Report on the Racial and Socio-Economic
Characteristics of Communities
with Hazardous Waste Sites**



COMMISSION FOR RACIAL JUSTICE
United Church of Christ
©1987

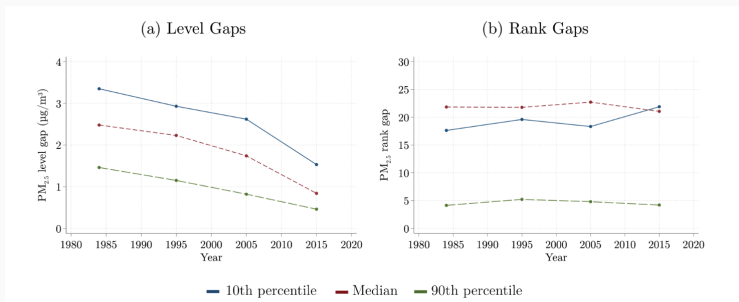
Potential Causes of Disparities

- Firm siting decisions: low land prices, low wages, not necessarily 'taste-based' discrimination
- 'Coming to the Nuisance': Sorting based explanations - access to jobs or environmental gentrification
 - Might be efficient *given existing distribution of resources*
- Coasian bargaining: If residents have property rights, may be able to hold up permitting processes
 - Low-income/minority communities have lower WTP for clean environment? Lower bargaining power? Again could be efficient *given existing distribution of resources*
 - Less secure property rights? Higher costs of enforcing property rights? (Language barriers, meeting locations, time costs)
- Political Economy: Minority communities may have less influence over institutional processes
 - Regulatory, monitoring/enforcement, remediation

Can Race Blind Policies Close Gaps?

Colmer et al: The Changing Nature of Pollution, Income, and Environmental Inequality in the United States - new data with individual level exposure

- The Clean Air Act in the US has improved air quality for black households more than white households
- Black HHs still face worse AQ than white HHs throughout the distribution



Takeaways

Existence of pollution exposure gaps is widely documented

- Environmental Impacts Frame: US Census data that tracks exposure at the individual and household level

Equity impacts of market-based instruments is an active area of research:

- Hernandez-Cortes and Meng (2023): Do environmental markets cause environmental injustice? Evidence from California's carbon market
- Deschenes and Weber: Equity Impacts of a Market for Clean Air

On the international stage, debates about who will pay for costs of mitigating and damages from climate change are central in current negotiations

- It all comes back to Coase: environmental issues are distributional issues